

Abhishek Bhattarai

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Software Engineer

Senior Software Engineer with **4+ years** of experience building scalable web, mobile, and backend applications used by millions of users. Specialized in **TypeScript (React, Next.js, Node.js)** and **Python**, with expertise in **full-stack architecture**, workflow automation, **AI-powered** developer tools, and high-performance execution platforms. Proven track record of delivering complex products across education technology and real estate, optimizing critical user experiences, automating engineering workflows, and collaborating across product, design, and engineering teams.

WORK EXPERIENCE

reAlpha Tech Corp. (Nasdaq: AIRE) • Ohio, USA (Remote) • 05/2026 - Present

Senior Software Engineer

Working on **Turnit**, a real estate turnover management platform used by property managers, vendors, and internal operations teams.

- Develop **end-to-end product** features across **web, backend**, and **mobile applications** using modern **TypeScript** technologies.
- Build **scalable** REST APIs, business workflows, and **database models** supporting real estate turnover operations.
- Develop responsive web interfaces and **cross-platform mobile** experiences while collaborating closely with product, design, and engineering teams.
- Own features throughout the complete software **development lifecycle**, from technical design and implementation to testing, deployment, and production support.
- Contribute to **system architecture**, code reviews, and engineering best practices to improve maintainability and delivery velocity.

Programiz • Lalitpur • 07/2022 - 04/2026

Senior Software Engineer

- **AI Strategy & Automation:** Architected a multi-stage AI content workflow using **GPT/Gemini** and **Slack API integrations**, automating the transition from raw **Documentation** to structured courses and increasing production velocity by **2x** from 12 to 6 days.
- **Lead Visualizer Architect:** Spearheaded the end-to-end engineering of the **Code Visualizer** suite (Python, C, and DSA); managed the architectural bridge between **TypeScript** state management and **Python Execution Pipelines** for 100K+ users.
- **Advanced Data Pipelines:** Designed a high-scale marketing data pipeline using **sGTM** and **Meta CAPI**, improving tracking accuracy by 50% and ensuring privacy-compliant **Conversion Tracking** while collaborating with **marketing and growth teams**.
- **System Performance:** Optimized critical user path, reducing authentication latency by **45%** through non-blocking background task re-architecture.
- **Mentorship:** Lead code reviews and technical **Documentation** standards, enforcing strict **TypeScript** type-safety and modern **REST API design** patterns.

Software Engineer

- **Execution Infrastructure:** Primary developer of the **Playground IDE**, a browser-based environment supporting 8 languages with real-time **Execution Pipelines** for 1M+ monthly learners.
- **Process Optimization:** Engineered a custom **Python/Node.js** Content Workflow System (CWS), reducing deployment cycles from 14 days to 4 days through **Workflow Automation**.
- **UX Reliability:** Debugged complex **cross-browser behavior** and frontend rendering bottlenecks, ensuring a seamless **Responsive UI** for learners on diverse devices and networks.
- **Feature Flag Systems:** Built the initial internal admin dashboard and feature flag controls, enabling granular product experimentation, operational visibility and easier software development and testing.

- **Scalable API Design:** Built and maintained **REST APIs** supporting Lessons and Quizzes, handling millions of requests per month with high reliability.
- Implemented **CI/CD pipelines and deployment automation**, improving reliability across environments.

EDUCATION

BSc (HONS) in Cyber Security and Ethical Hacking

Coventry University • GPA: 3.6+ • 09/2019 - 09/2022

SKILLS

Core Tech Stack: TypeScript, Next.js, React, Node.js, Python, PostgreSQL, React Native, Tailwind CSS

Languages: SQL (MySQL/PostgreSQL/BigQuery), Rust (Exploratory/Personal projects), JavaScript, HTML / CSS

Frontend Architecture: Design Systems, Component Libraries, Responsive UI, Cross-browser behavior optimization, Framer Motion

Backend & Systems: REST API design, Microservices, Execution Pipelines, Docker, CI/CD, Workflow Automation, AI Pipeline Design

Growth & Marketing Analytics: Google Tag Manager (GTM), Server-side GTM (sGTM), Meta Pixel, Facebook CAPI (Conversions API), GA4, Conversion Tracking, Funnel Analysis, RudderStack, Mixpanel

PROJECTS

Turnit

reAlpha

Technologies: TypeScript, React, Next.js, React Native, Node.js, PostgreSQL

- Full-stack contributor to a real estate turnover management platform used by property managers, vendors, and operations teams.
- Develop end-to-end features across web, backend, and mobile applications.
- Design scalable APIs, business workflows, and database models supporting property turnover operations.
- Collaborate across product, design, and engineering to deliver production-ready features.

Code Visualizer Suite (Python, C, DSA)

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Technologies: React, TypeScript, Python, Node.js

- Led engineering of interactive Python, C, and DSA visualizers serving 100K+ monthly learners.
- Designed backend execution services, React/TypeScript frontend architecture, debugger state extraction, recursive call visualization, and deployment pipelines.
- Built highly interactive algorithm visualizations with optimized rendering performance and intuitive user experiences.

Playground (Browser-based IDE)

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Technologies: React, TypeScript, Node.js

- Developed a browser-based IDE supporting eight programming languages and serving over one million monthly learners.
- Built backend services and scalable APIs powering real-time code execution.
- Improved platform reliability and developer experience through performance optimizations and deployment improvements.

AI-Powered Content Engine

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Technologies: Python, GPT, Gemini, Slack API

- Architected an AI-powered content generation platform integrating GPT, Gemini, and Slack.
- Automated course creation workflows, reducing production time from 12 days to 6 days.
- Enabled scalable content generation while maintaining editorial consistency.

Guided Project Environment

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Analytics Tracking & Marketing Data Pipeline

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CERTIFICATIONS

Applied Data Science Labs • 08/2023 - 08/2023

World Quant University

VOLUNTEERING & LEADERSHIP

Edulift Organization • 12/2025 - 12/2025

Resource Person/ Trainer

Led an interactive training for 18 government schoolteachers, teaching how to create engaging digital lessons, use quizzes for instant feedback, and apply Google Classroom for announcements and assignments.